

Low-grade heat utilization in the methanol-fired gas turbines through a thermochemical fuel transformation

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ABSTRACT

Methanol can be considered as an alternative fuel for different engines. This paper investigates the low-grade heat utilization in methanol-fired gas turbines. Two cases were analyzed: the first is low-grade heat utilization for steam generation, and the second is for methanol decomposition before its combustion. The concept of low-grade heat utilization via methanol decomposition is developed and described in details. The comparative thermodynamic analysis was performed via Aspen HYSYS under the wide ranges of the operational parameters: gas turbine inlet temperature of 800–1000 °C, compression/expansion ratio of 9–15. The temperature potential of low-grade heat was set from 200 to 400 °C. It was established that low-grade heat utilization via methanol decomposition is more efficient (up to 5%) in comparison to low-grade heat utilization via steam generation. Methanol decomposition before its combustion leads to an increase in the power output in the gas turbine. Simulation tests of various temperatures potential of low-grade heat showed that it does not play a significant role in the efficiency of utilization. However, it was determined that maximal methanol decomposition is observed at a temperature above 300 °C. The Sankey diagrams for the energy flows are depicted and analyzed. The future perspectives of the use of the low-grade heat utilization concept via fuel decomposition were described for carbon-free fuel-ammonia.

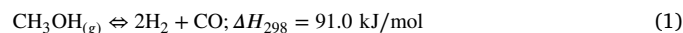
1. Introduction

The development of modern society is inextricably linked with energy engineering. The main trend in recent years for the development of energy engineering is low-carbon energy generation [1]. The utilization of renewable energy sources such as solar and wind energy for electricity generation did a big step in the last years [2]. Moreover, the development of organic Rankine cycles made it possible for the efficient utilization of low-grade heat (up to 300 °C) for electricity generation [3,4]. However, despite of successes of renewable energy, the share of total electricity generation from hydrocarbon fuel is more than 60% [5]. Therefore, one of the main ways of energy development is the use of hydrocarbon fuels with low or zero carbon footprint. Methanol is one of these fuels [6,7]. A conventional way of methanol production is hydrocarbon fuel transformation. However, the last years showed that methanol can be efficiently produced from renewable sources, for example, biomass with a highly low carbon footprint [8,9].

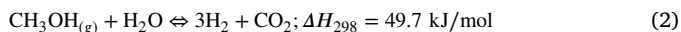
Renewable methanol can be used as a fuel in the traditional power plants, marine, and automotive sectors, etc. Verhelst et al. provided a comprehensive review of the methanol utilization as a pure fuel or blend component for internal combustion engines [10]. Svanberg et al.

presented the review on the use of renewable methanol for maritime shipping [11]. Gupta et al. published an overview on the use of bio-methanol as a fuel in the gas turbine units [12]. Ayaz et al. reported the perspectives of bio-methanol for a micro gas turbine for sustainable electricity generation [13]. The above-mentioned reviews are showing that methanol is a technically viable option to use in various fuel-consuming equipment and there are no major challenges with the potential replacing of the traditional hydrocarbon fuels with methanol, including bio-methanol.

The above-mentioned examples are showing that renewable methanol can be efficiently used in various fuel-consuming equipment. One of the main disadvantages of methanol as a fuel is the low heat of combustion. The lower heat value of methanol is 19.930 MJ/kg which is notable less than it is for natural gas or diesel. On the overhand, methanol can be easily converted to synthesis gas (hydrogen-rich fuel). Methanol decomposition and methanol steam reforming reactions are the main processes for the transformation of methanol to synthesis gas [14].



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The reactions of methanol transformation to hydrogen-rich gas are endothermic; thus, the combustion heat of the reaction products (in terms of 1 mole of methanol) is higher than it is for methanol. Fuel reforming before combustion is widely discussed as a way of the use of exhaust heat [15–19]. Tartakovsky and Sheintuch provided a comprehensive overview on fuel reforming in internal combustion engines [20]. In their paper, a historical retrospective of fuel reforming before combustion was provided. Lee and Wimmer [21] firstly reported that fuel reforming before combustion provides better emission characteristics of the combustion process. The scientific group led by Prof. Tartakovsky published a set of papers on the investigation of the effect of fuel reforming before combustion on emission and efficiency performance of ICEs [22–26]. The main advantages of such a way of methanol utilization are the possibility to use exhaust heat for fuel reforming, and the combustion of hydrogen-rich gas as a fuel. Therefore, better efficiency and combustion performances are taking place when methanol is reformed before combustion in ICEs. Poran and Tartakovsky showed that an efficiency of ICE with methanol reforming is 12%–14% higher than an efficiency of gasoline-fed engine [22]. Nguyen et al. investigated an reformed-exhaust gas recirculation concept to improve the efficiency of methanol fired ICE [27]. Authors showed that an thermal efficiency of ICE with methanol reforming is 7.11% higher than an efficiency of ICE with direct methanol combustion.

Wang et al. reported the results of an investigation of thermochemical recuperation in a methanol-fueled gas turbine [28]. Authors showed that methanol conversion is achieved up to 99.6% when exhaust heat is used as a heat source to cover an endothermic effect. Pettersson and Sjöström provided the review on the use of the products of methanol decomposition in the gas turbine units through exhaust heat recuperation [29]. Authors reported that an overall thermal efficiency of gas turbine with methanol fuel reforming is 16% higher than that of an ordinary gas turbine. Moreover, Tola and Lonis presented a paper on the use of bio-methanol in a gas turbine with thermochemical recuperation [30]. The main attention in this paper was paid to the fact that renewable methanol can be efficiently used for reforming using low-grade exhaust heat, and then the reforming products are used as a fuel. The authors concluded that methanol-fueled gas turbine with fuel reforming before combustion shows very low CO₂ emissions. Tola and Lonis showed that a gas turbine efficiency with methanol fuel reforming is up to 6%–8% higher than that for a gas turbine with direct methanol combustion.

Methanol decomposition has some advantages in comparison to methanol steam reforming. The main of it is no need to produce steam for reforming. There is sufficient information from the published papers that over Ni, Pt, and Cu-based catalysts methanol can be converted to a mixture of hydrogen and carbon monoxide with negligible low fractions of other species, e.g. methane, steam, carbon dioxide [31–33]. Moreover, the methanol decomposition reaction can proceed at the moderate low temperatures (up to 300 °C). Therefore, low-grade heat can be efficiently used as a heat source for the methanol decomposition reaction.

The traditional way of the use of low-grade heat is steam or organic Rankine cycle. However, according to the first law of thermodynamics, the efficiency of transformation of heat to electricity is strongly depending on the hot temperature reservoir. Therefore, due to the low temperature of low-grade heat sources the efficiency of use is quite low. One of the ways to improve the efficiency of low-grade heat transformation to electricity is utilization in the Brayton cycle fired with methanol. However, the temperature of the hot temperature reservoir in such cycles can be reached up to 1200–1500 °C. How can low-grade heat be used in the methanol-fired Brayton cycle? The answer is the following. Low-grade heat can be used for the endothermic transformation of

methanol to hydrogen-rich fuel. In this case, the low-grade heat will be converted into chemical energy of the new hydrogen-rich fuel, and then this new fuel will be burned in a combustion chamber. In other words, low-grade heat will be converted to high-grade heat. The system “methanol – decomposition process with use of low-grade heat – hydrogen-rich fuel” can be considered as “chemical exergy pump”.

Therefore, the main goal of this paper is to provide the performance evaluation of the methanol-fueled gas turbine with two configurations: the first is direct methanol combustion and the second is endothermic methanol transformation into hydrogen-rich fuel and combustion of the products of transformation. Moreover, the efficiency of conversion of low-grade heat to electricity for both configurations was determined. Insofar as the efficiency of the methanol decomposition process is strongly depending on temperature and pressure, the performance evaluation was conducted under a wide range of operating conditions.

2. Low-grade heat utilization concept

The main concept of low-grade heat utilization in a methanol-fueled gas turbine is the use of this heat for the endothermic methanol decomposition via reaction (1). To show this concept, two schematic diagrams of the gas turbine with a bottom Rankine cycle are presented in Fig. 1. Fig. 1a shows the traditional way of low-grade heat utilization via steam generation for a Rankine cycle. In this scheme, low-grade heat is used to produce steam, and steam is used in a steam turbine. Thus, in the scheme from Fig. 1a low-grade heat is used in a steam turbine cycle.

Fig. 1b shows the gas turbine with preliminary methanol decomposition before combustion with the utilization of low-grade heat. In this scheme, low-grade heat is utilized for the endothermic process of methanol decomposition. As the result of this decomposition, hydrogen-rich fuel is produced that is then burned in a combustion chamber. Therefore, in this scheme, low-grade heat is used in a Brayton cycle that has a notable higher temperature level than a Rankine cycle.

Both schemes in Fig. 1 contain the bottom Rankine cycles because the temperature of the exhaust gas after the gas turbines is moderately high and can be used also for steam generation for a steam turbine cycle. A comparison of the efficiency of these schemes will show the benefits of the use of low-grade heat for the endothermic transformation of methanol.

Obviously, the electricity generated in the gas turbine cycle in the scheme (b) is higher than it for scheme (a). In parallel, the electricity generated in the steam turbine cycle in the scheme (a) is higher than it in scheme (b). However, according to the first law of thermodynamic, the cycle efficiency is higher when the cycle temperature is higher. Therefore, it should be expected that low-grade heat is used more efficiently in the scheme with methanol decomposition than in the conventional steam turbine cycle with using low-grade heat for steam generation.

In the scheme from Fig. 1a, the conversion of chemical energy of methanol into heat is proceeding in one stage by direct methanol combustion. In the scheme from Fig. 1b, the conversion of chemical energy of methanol into heat is divided into two stages. The first stage is the endothermic methanol decomposition by utilization of low-grade heat. The second stage is the combustion of hydrogen-rich gas produced in the methanol decomposition process. Schematically, the conversion of chemical energy of methanol into heat for the schemes (a) and (b) from Fig. 1 is presented in Fig. 2.

3. Methodology

3.1. Cycle analysis

The thermal efficiency of power plants is the main criterion of effectiveness. The thermal efficiency of a combined cycle with the utilization of low-grade heat can be determined as follows:

$$\eta_{CC} = \frac{N_{CC}}{\dot{m}_{\text{MeOH}} \cdot LHV_{\text{MeOH}} + H_g} \quad (3)$$

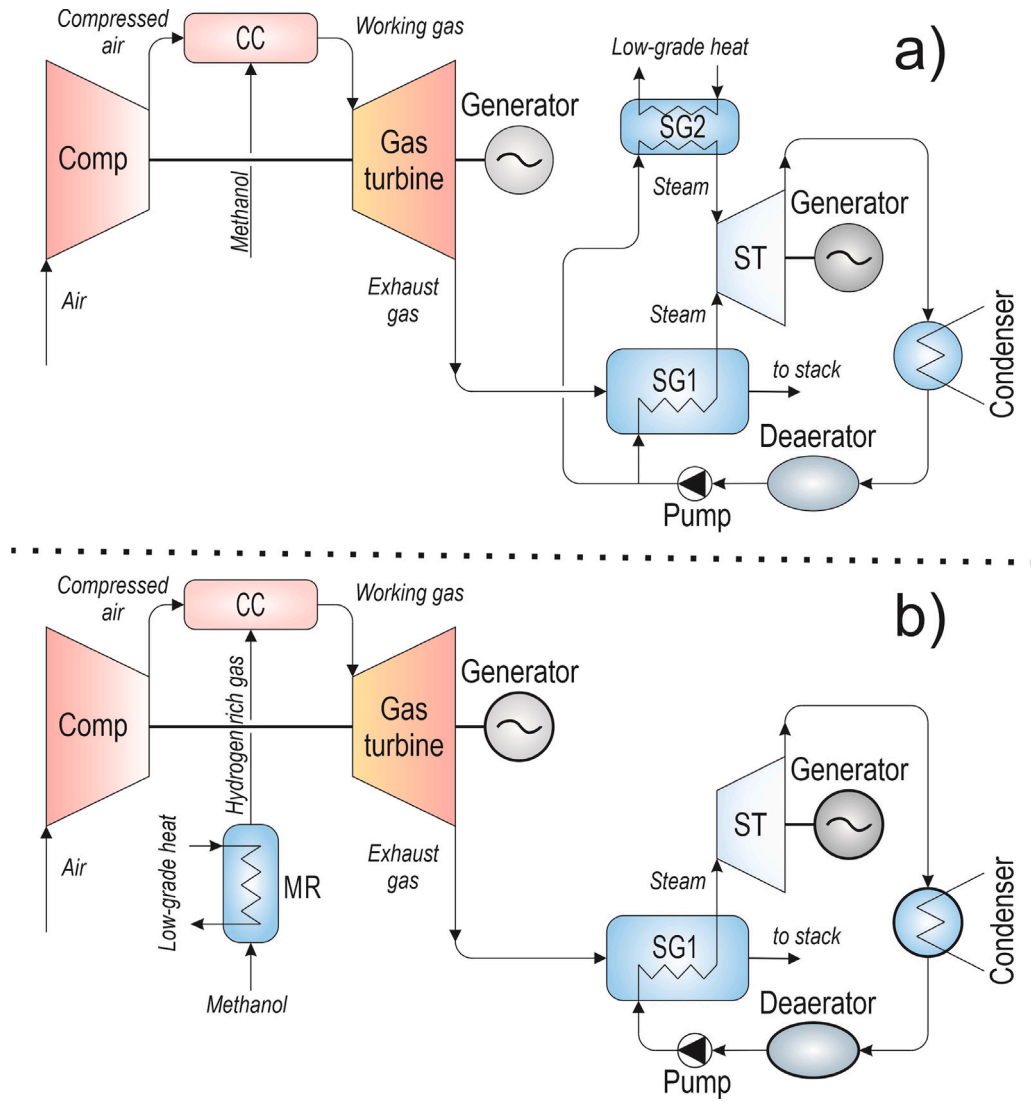


Fig. 1. The schematic diagrams of the gas turbines with a bottom Rankine cycle fueled with methanol: (a) direct methanol combustion; (b) methanol decomposition and hydrogen rich gas combustion: CC — combustion chamber; MR — methanol reformer; SG — steam generator; ST — steam turbine.

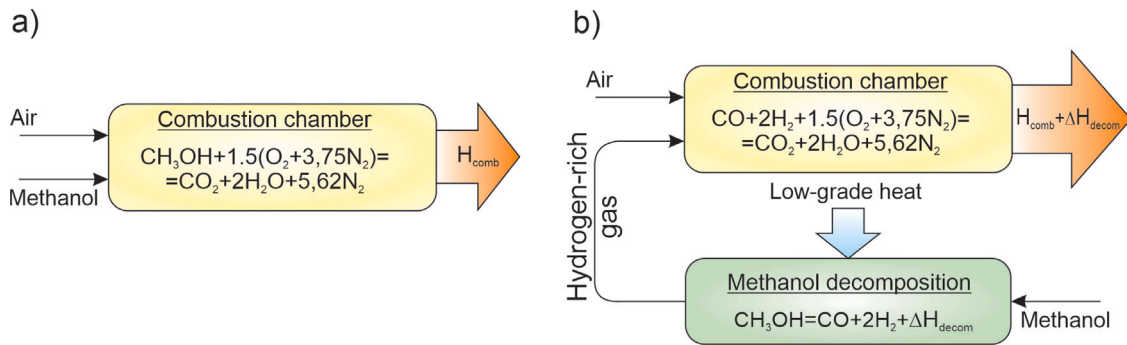


Fig. 2. Conversion of chemical energy of methanol into heat for the schemes (a) and (b) from Fig. 1.

where N_{CC} is output power of a combined cycle; $\dot{m}_{MeOH}$ is the mass flow rate of methanol fed to the cycle; LHV_{MeOH} is the lower heat value of methanol; H_{lg} is enthalpy of utilized low-grade heat.

The schematic diagrams from Fig. 1 consist of two parts: the gas turbine cycle and the steam turbine cycle. Therefore, the overall output power can be calculated as a sum of whole power inputs and outputs

as follows:

$$N_{CC} = (N_{GT} - N_C) + (N_{ST} - N_P) \quad (4)$$

where N_{GT} , N_{ST} is the power of a gas turbine and steam turbine, respectively; N_C is the power of air compressor, respectively; N_P is the power of a pump.

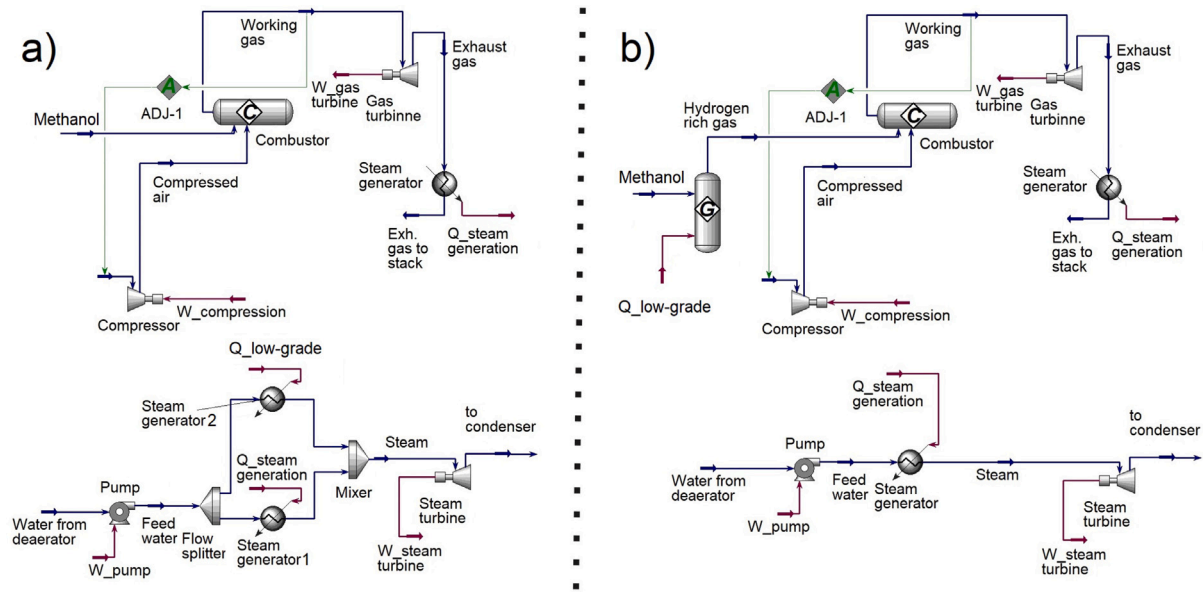


Fig. 3. Computational scheme of the combined cycles with low-grade utilization in Aspen HYSYS: (a) for steam generation; (b) for methanol decomposition.

The first term of Eq. (4) shows the net output power of a gas turbine cycle, the second term presents the net output power of a steam turbine cycle.

The main difference between a combined cycle with low-grade heat utilization for methanol decomposition and a conventional combined cycle system using low-grade heat for steam generation is a pathway of low-grade heat utilization. In a combined cycle with methanol decomposition, low-grade heat is used in a gas turbine cycle, while in a conventional combined cycle low-grade heat is used in a steam turbine cycle. Therefore, the efficiency of gas turbine and steam turbine cycles for the schemes (a) and (b) from Fig. 1 can be determined by the following relationships:

$$(a) \eta_{GT} = \frac{N_{GT}}{\dot{m}_{MeOH} \cdot LHV_{MeOH}}; \quad (b) \eta_{GT} = \frac{N_{GT}}{\dot{m}_{MeOH} \cdot LHV_{MeOH} + H_{lg}}; \quad (5)$$

$$(a) \eta_{ST} = \frac{N_{ST}}{H_{exh} + H_{lg}}; \quad (b) \eta_{ST} = \frac{N_{ST}}{H_{exh}}; \quad (6)$$

where subscriptions a, b are corresponding to the schemes from Fig. 1, respectively; N_{GT}, N_{ST} are power output of the gas turbine and steam turbine cycles, respectively; H_{exh}, H_{solar} is the heat flows with exhaust gas and low-grade heat.

The thermodynamic analyses of both cycles from Fig. 1 have been conducted via Aspen HYSYS [34–36].

The computational schemes for the schemes from Fig. 1a in Aspen HYSYS are presented in Fig. 3. It should be noted that in all analyzed cases, the heat flows of low-grade heat were equal for scheme (a) and scheme (b).

The schemes were analyzed for various gas turbine inlet temperatures and pressures (expansion ratios). The regulation of turbine inlet temperature was performed via changes in the mass flow rate of air. For this regulation, function (Adjust) was used to set the accurate mass flow rate. Low-grade heat in the computational scheme is spent for the methanol evaporation and for the endothermic reaction of methanol decomposition:

$$H_{lg} = H_{evap} + \Delta H_{298} \quad (7)$$

where $H_{evap}, \Delta H_{298}$ are heat flows for the methanol evaporation, and the endothermic effect of the methanol decomposition process, respectively.

3.2. Methanol decomposition

Among two ways of methanol reforming – methanol decomposition (Eq. (1)) and methanol steam reforming (Eq. (2)) – the first one is more capable for using in the schematic diagram from Fig. 1b. In this case, there is no steam generation for methanol reforming. In the case when the methanol steam reforming process is taking place for thermochemical fuel transformation, steam from the reaction mixture is converted to synthesis gas and then this synthesis gas is converted to flue gas. Thus, flue gas is containing more steam in the flue gases in comparison with direct methanol combustion. Therefore, heat flow for steam generation from liquid water (latent heat) is rejected to the atmosphere and heat flow for water evaporation becomes exhaust heat. The flue gas composition for direct methanol combustion and combustion of the product of methanol decomposition is the same.

The methanol decomposition process is a well-studied process both theoretically and experimentally [37–39]. Aspen HYSYS is compatible to perform the thermodynamic analysis of methanol decomposition. The main goal of such analysis is a determination of heat and mass flows in a methanol reformer. Methanol is decomposed to produce hydrogen, and carbon monoxide, as well as the decomposition products, may contain unreacted methanol. Therefore, in this analysis, it was assumed that the decomposition products consist of H_2 , CO , CH_3OH .

The equilibrium composition of the reforming products was calculated by using the Gibbs free energy minimization method. The Peng–Robinson equation of state is used for this method. Previously, an author published a set of papers on thermodynamic analysis of the steam methane reforming process and a detailed algorithm is described in these papers [40,41]. The thermodynamic analysis of the steam methane reforming process was employed via Aspen HYSYS.

Moreover, the thermodynamic analysis makes it possible to determine the heat flows in the methanol reformer. Whole low-grade heat is fed to evaporate methanol, to heat methanol, and to cover the endothermic effect of the methanol decomposition process.

3.3. Model setup

The thermodynamic analyses were conducted for various operating parameters such as temperature and pressure of methanol decomposition. Stream properties and efficiency parameters of the gas and steam turbine cycles are presented in Table 1.

On further analysis, the following assumptions were applied:

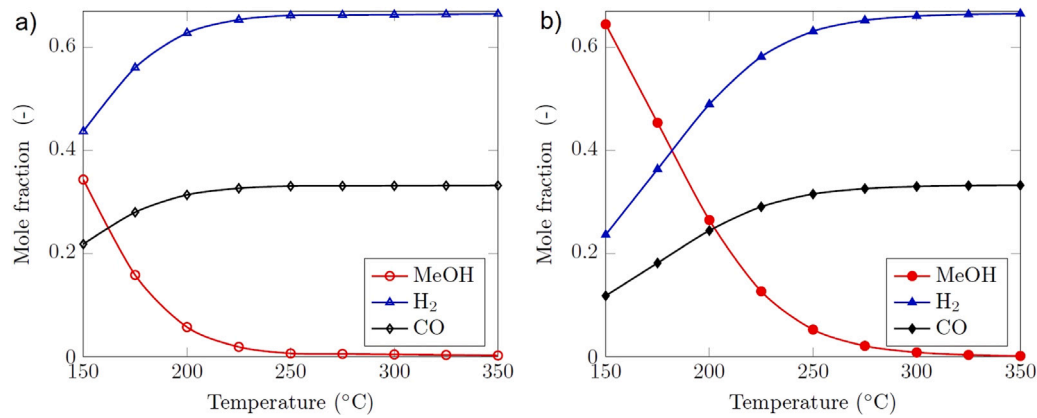


Fig. 4. Composition of methanol decomposition products as a function of temperature for pressure of 5 bar (a) and 15 bar (b).

Table 1

Properties of streams and efficiency parameters of the gas and steam turbine cycles.

Parameter	Value
CH ₃ OH mass flow rate	100 kg/h
Gas turbine inlet temperature (TIT)	1000; 900; 800 °C
Compression/expansion ratio	15; 12; 9
Pressure losses	0
Turbine isentropic efficiency	90%
Compressors isentropic efficiency	85%
Steam temperature at steam turbine inlet	300 °C
Steam pressure at steam turbine inlet	15 bar
Condenser pressure	0.06 bar
Temperature of atmospheric air and methanol	20 °C
Temperature of methanol decomposition	400; 300; 200 °C

- no pressure losses in all elements of the schematic diagrams;
- no losses in a combustion chamber;
- pressure of the decomposition process is equal to pressure of air after compressor for each case.

4. Results and discussion

4.1. Steam methane reforming

The equilibrium composition of the products of methanol decomposition is depending on temperature and pressure according to Le-Chatelier's principles. An increase in temperature in methanol decomposition. On the other hand, an increase in pressure leads to a decrease in methanol decomposition. The effect of temperature and pressure on the composition of methanol decomposition products is presented in Fig. 4. Usually, the methanol decomposition process operates in the temperature range up to 350–400 °C. This temperature range is corresponding to the temperature of low-grade heat.

Le-Chatelier's principles are well illustrated by the effect of temperature and pressure on the decomposition process. Fig. 4 shows that methanol is almost fully reacted in the temperature range above 250 °C for 5 bar, and in the temperature range above 300 °C for 15 bar. According to Le-Chatelier's principle, the system will oppose changes in the system to achieve a new equilibrium state. Therefore, increase in temperature leads to increase in methanol decomposition, while increase in pressure leads to decrease methanol decomposition.

Operational parameters such as temperature and pressure affects also on heat balance in the reformer. Fig. 5 shows the heat flow from low-grade heat in the reformer. The total heat flow for the methanol decomposition process is virtually divided into three parts: methanol evaporation, methanol heating, and methanol decomposition. Fig. 5 depicts that an increase in temperature leads to an increase in heat flow to cover an endothermic effect of the methanol decomposition reaction.

Heat flow for methanol evaporation and heating is slightly increasing with an increase in temperature. Based on Fig. 5, it can be concluded that the appropriate temperature for low-grade utilization via methanol decomposition is the temperature range above 250 °C. In this case, the maximum endothermic effect of the methanol decomposition reaction is observed.

4.2. Cycle analysis

To understand the effect of the operational parameters on the efficiency of low-grade heat utilization, thermodynamic analysis was performed for a wide range of operational parameters. Fig. 6 shows the effect of operational parameters on the efficiency of a combined cycle with low-grade heat utilization for two ways of low-grade utilization corresponding to the schemes from Fig. 1: low-grade heat is used for steam generation for a steam turbine and low-grade heat is used for methanol decomposition and utilization of the products of decomposition in a gas turbine. Fig. 6a depicts a comparison of the efficiency for a combined cycle with methanol decomposition (MD) - dashed lines, and without methanol decomposition - solid lines. Fig. 6 presents a comparison of the efficiency for a combined cycle at various temperatures of low-grade heat: solid lines - gas turbine inlet temperature of 1000 °C; dashed lines - gas turbine inlet temperature of 800 °C.

Fig. 6a shows that utilization of low-grade heat for methanol decomposition leads to an increase in the efficiency up to 5% in comparison with the cycle where low-grade heat is used for steam generation. The efficiency of a combined cycle increases with an increase in gas turbine inlet temperature. For example, at a compression ratio of 15 (pressure of 15 bar) the efficiency of a combined cycle at TIT of 1000 °C with methanol decomposition is 49.83% while the efficiency without methanol decomposition is 44.78% - the difference is 5.15%. At the same time, at a compression ratio of 15 (pressure of 15 bar) and TIT of 800 °C, the efficiency with MD is 45.39% and without MD is 41.33% - the difference is 4.06%. Therefore, for a high temperature of inlet working fluids, the difference between a combined cycle with MD and without MD is higher. This fact can be explained by the first law of thermodynamics. An increase in temperature leads to an increase in efficiency. In these cases, an increase in temperature leads to an increase in the efficiency of a combined cycle, and low-grade heat is utilized in the cycle with higher temperatures. At the same time, utilization of low-grade heat in the cycle with steam generation leads to an increase in the total power output from a steam turbine cycle which is operated under lower efficiency because the temperature of steam is lower than the temperature of working fluid in a gas turbine.

Based on the Eqs. (5) and (6), the efficiency of a gas turbine and steam turbine cycle was determined. Table 2 shows the efficiency of a gas turbine cycle and a steam turbine cycle.

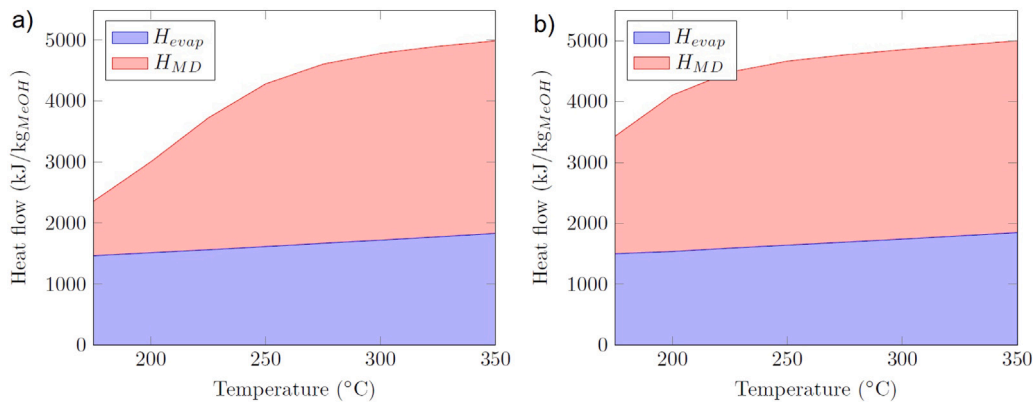


Fig. 5. Heat flows in the reformer for methanol evaporation and heating (H_{evap}), and methanol decomposition (H_{MD}).

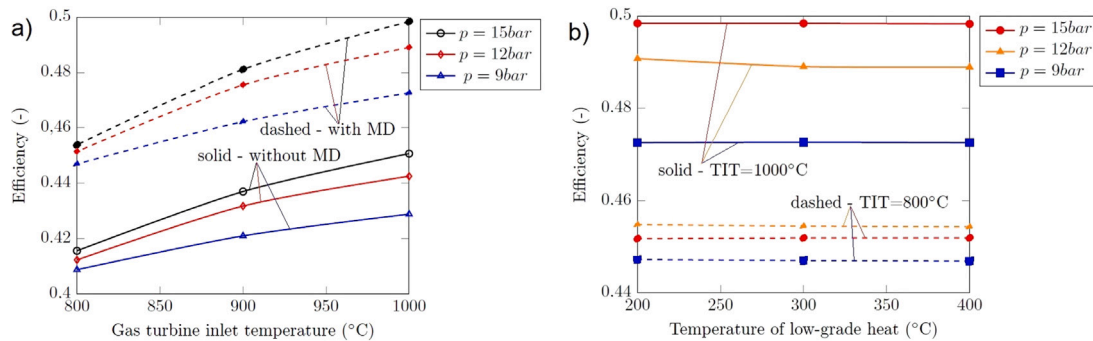


Fig. 6. Efficiency of a combined cycle with low-grade heat utilization under various operating conditions: (a) comparison of the efficiency for a combined cycle with methanol decomposition (MD) — dashed lines, and without methanol decomposition — solid lines; (b) comparison of the efficiency for a combined cycle at various temperature of low-grade heat: solid lines — gas turbine inlet temperature of 1000 °C; dashed lines — gas turbine inlet temperature of 800 °C.

Table 2
Comparison of the efficiency of a gas turbine cycle and steam turbine cycle.

TIT, °C	15 bar			12 bar			9 bar		
	1000	900	800	1000	900	800	1000	900	800
Gas turbine cycle efficiency, %	37.23	35.72	33.28	35.76	34.62	32.83	33.36	32.57	31.36
Steam turbine cycle efficiency, %	24.68	24.68	24.68	24.68	24.68	24.68	24.68	24.68	24.68

Moreover, Fig. 6b shows the effect of low-grade heat temperature on the efficiency of a combined cycle. It is clearly seen that the temperature of low-grade heat does not have a notable effect on the efficiency of a combined cycle. Therefore, the low-grade heat with a temperature of even 200 °C can be efficiently utilized for electricity generation with efficiency up to 50% when methanol decomposition takes place. The temperature of low-grade heat affects on amount of low-grade heat which is used in the gas turbine cycle. An increase in the low-grade heat temperature leads to increase in a heat flow in a reformer, and as a result decreasing in a primary methanol consumption.

To better understand the heat flows in a combined cycle with low-grade heat utilization, the Sankey diagrams for the schemes from Fig. 1 are presented in Fig. 7. Fig. 7a is corresponding to the scheme from Fig. 1a – low-grade heat utilization for steam generation; Fig. 7b is corresponding to the scheme from Fig. 1b – low-grade heat utilization via methanol decomposition. Fig. 7 shows that utilization of low-grade heat via methanol decomposition leads to an increase in power output from the gas turbine cycle. At the same time, utilization of low-grade heat by steam generation leads to an increase in the power output of the gas turbine cycle. However, according to Table 2, the efficiency of the gas turbine cycle is higher than the efficiency of the steam turbine cycle. Therefore, the utilization of low-grade heat via methanol decomposition is more efficient in comparison to the traditional way

via steam generation for the steam turbine. Fig. 7 shows that exhaust heat after a gas turbine is higher for the scheme in Fig. 1b than for the scheme Fig. 1a. The reason for it is an increase in the mass flow rate of compressed air. As a result, the mass flow rate of the working fluid in the scheme Fig. 1b is higher than it for the scheme Fig. 1a.

Fig. 7 shows that for the constant energy flow with methanol of 563.6 kW (corresponding to methanol mass flow rate of 100 kg/h) the energy flows in the steam turbine and gas turbine generators are different. In the power plant with direct methanol combustion the power of steam and gas turbine is 107.5 kW and 221.4 kW, respectively (total power is 328.9 kW). At the same time, for the power plant with methanol decomposition and combustion of reforming products, the power of steam and gas turbines is equal to 86.4 kW and 255.7 kW, respectively (total power is 342.1 kW). Therefore, for the constant mass flow rate of methanol, the total power of the power plant is 13.2 kW higher for the power plant with methanol decomposition.

4.3. Future perspectives

The above-mentioned example showed that methanol thermochemical decomposition can be considered as an efficient way of low-grade heat utilization. Methanol is one of the promising biofuels in the near future. However, even though methanol can be produced from biomass, the combustion of methanol or its decomposition products

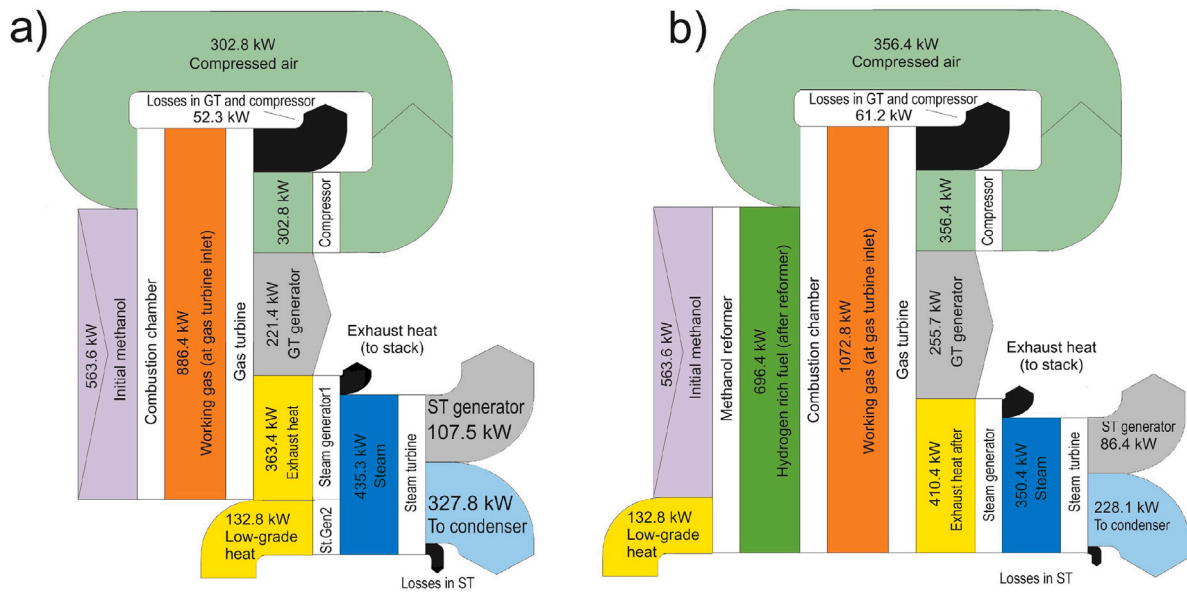


Fig. 7. Sankey diagram for the combined cycles with utilization of low-grade heat via (a) steam generation for a steam turbine and (b) decomposition of methanol before combustion: gas turbine inlet temperature of 1500 °C; compression/expansion ratio of 15 (pressure in the reformer of 15 bar).

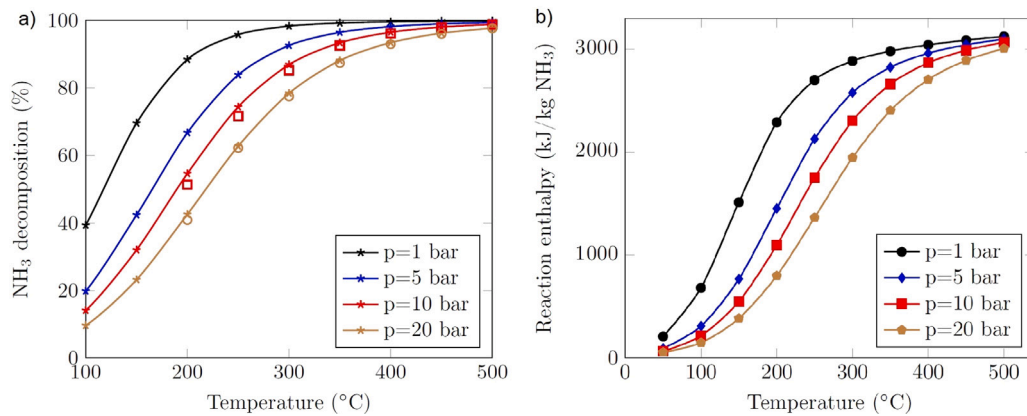
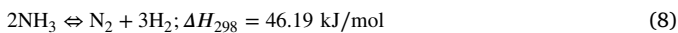


Fig. 8. Ammonia decomposition for various temperature and pressure: hollow points — experimental results [42]; solid point — equilibrium state.

is still producing carbon dioxide. One of the carbon-free fuels for the implementation of low-grade heat utilization is ammonia. The process of thermochemical ammonia decomposition is well studied and the chemical reaction is as follows:



Therefore, it is important to note that ammonia-fired gas turbines can be efficiently integrated with low-grade heat utilization. Fig. 8 shows the characteristics of the ammonia decomposition process determined in the previous author's study [41]. Fig. 8 shows that the temperature level of low-grade heat for full ammonia decomposition has to be slightly higher than it does for methanol decomposition (for a given pressure). For example, at pressure 5 bar, the methanol conversion of above 99% is observed in the temperature range above 257 °C, while the ammonia conversion above 99% is taking place in the temperature range above 412 °C. The detailed investigation of ammonia-fired gas turbine with low-grade heat utilization via ammonia decomposition is a promising topic for study and industrial application.

Among various thermochemical processes that can be applied for low-grade heat utilization, the decomposition processes are more favorable than the reforming reactions with oxidizers, e.g. steam. This is because steam for reforming is converted to synthesis gas and then to flue gas. Additional steam from flue gases is injected into the

atmosphere. Therefore, the latent heat for steam generation from water became the heat losses [43]. The steam reforming processes can have efficient applications if additional steam from the flue gases will be condensed and latent heat of condensation will be usefully used in the fuel cycle.

5. Conclusion

In this paper, the thermodynamic analysis of a low-grade heat utilization concept through a thermochemical decomposition of methanol in a gas turbine was performed via Aspen HYSYS. It is established that utilization of low-grade heat for methanol decomposition leads to an increase in the efficiency up to 5% in comparison with the cycle where low-grade heat is used for steam generation. The efficiency of a combined cycle increases with an increase in gas turbine inlet temperature. For example, at a compression ratio of 15 (pressure of 15 bar) the efficiency of a combined cycle at TIT of 1000 °C with methanol decomposition is 49.83% while the efficiency without methanol decomposition is 44.78% – the difference is 5.15%. At the same time, at a compression ratio of 15 (pressure of 15 bar) and TIT of 800 °C, the efficiency with MD is 45.39% and without MD is 41.33% – the difference is 4.06%. Therefore, for a high temperature of inlet working fluids, the difference between a combined cycle with

MD and without MD is higher. It was established that the temperature potential of low-grade heat does not notably affect the efficiency of a combined cycle. Therefore, the low-grade heat with a temperature of even 200 °C can be efficiently utilized for electricity generation with efficiency up to 50% when methanol decomposition takes place. The Sankey diagrams of the energy flows showed that utilization of low-grade heat via methanol decomposition leads to an increase in power output from the gas turbine cycle. At the same time, utilization of low-grade heat by steam generation leads to an increase in the power output of the gas turbine cycle.

The perspectives of low-grade heat utilization through ammonia decomposition are briefly considered. It is shown that endothermic fuel decomposition via low-grade heat utilization can be considered as a “chemical exergy pump”.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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